

Latest Biomarkers for Alzheimer's Disease

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Executive Summary

1. Fluid Biomarkers (Blood and CSF)

Core Established Biomarkers

- **Amyloid- β ($A\beta_{42}$ and $A\beta_{42}/40$ ratio):** Measured in both CSF and plasma; reflects amyloid plaque pathology. The plasma $A\beta_{42}/40$ ratio has emerged as a particularly accessible screening tool.
- **Phosphorylated tau (P-tau):** Multiple species (P-tau181, P-tau217, P-tau231) measured in CSF and increasingly in blood; reflects tau pathology and AD-specific changes. P-tau217 is often highlighted as among the most promising blood-based biomarkers.
- **Total tau (T-tau):** Nonspecific marker of neurodegeneration, traditionally measured in CSF.

Blood-Based Biomarkers (Most Rapidly Developing Area)

- **Plasma $A\beta_{42}/40$ ratio:** Cost-effective and scalable; considered likely to "revolutionize" diagnostic workup globally.
- **Plasma P-tau217:** Strong correlation with CSF and PET findings; shows high accuracy for AD detection.

- **Neurofilament light chain (NfL):** Blood-based marker of general neurodegeneration; nonspecific but useful for monitoring axonal injury.
- **Glial fibrillary acidic protein (GFAP):** Marker of astroglial activation/reactive astrogliosis.
- **Synaptic proteins:** Neurogranin (CSF), which appears AD-specific and predicts rate of cognitive deterioration.

Emerging Biomarkers

- **Mitochondrial microRNAs (miRNAs):** Regulate synaptic and mitochondrial function; differentially expressed in AD.
- **DNA methylation profiles:** Epigenetic signatures associated with AD progression.
- **Long non-coding RNAs (lncRNAs):** Investigated as diagnostic biomarkers and therapeutic targets.
- **Inflammatory/neuroimmune markers:** Reflect neuroinflammation component of AD.

2. Imaging Biomarkers

PET Imaging

- **Amyloid PET** (e.g., florbetapir, florbetaben): Detects fibrillar amyloid plaques; used for eligibility for anti-amyloid therapies and monitoring treatment response.
- **Tau PET** (e.g., flortaucipir): Detects neurofibrillary tangle pathology; enables region-specific staging.
- **FDG-PET:** Metabolic marker for synaptic dysfunction; useful in atypical AD variants (dysexecutive, behavioral variant, posterior cortical atrophy, logopenic PPA).

MRI

- **Structural MRI:** Detects atrophy patterns (hippocampal, posterior cortical, frontal variants).
- **Functional MRI:** Network-level changes.

Updated A/T/N Framework

The original amyloid/tau/neurodegeneration classification has been expanded to include:

- **A:** Amyloid pathology
- **T:** Tau pathology
- **N:** Neurodegeneration
- **I:** Neuroinflammation (newer addition)
- Additional pathologic targets are being incorporated for biomarker detection.

3. Consensus and Key Trends

- **Shift toward blood-based biomarkers:** Multiple sources emphasize that plasma-based tests (A β 42/40 ratio, P-tau217, NfL, GFAP) represent the most significant recent advance due to scalability and cost-effectiveness.
- **Earlier diagnosis focus:** Biomarker development is driven by the need to identify AD in preclinical/presymptomatic stages, which is essential given new disease-modifying therapies (lecanemab, donanemab).
- **Multimodal approach:** Combination of fluid and imaging biomarkers enables more precise pathological staging and individualized treatment plans.
- **Pathological staging:** Biomarkers now enable staging of AD across the continuum from preclinical/asymptomatic to severe dementia.

4. Areas of Active Investigation/Uncertainty

- **Optimal blood biomarker combinations:** Whether plasma A β 42, A β 42/40 ratio, NfL, or a combination will become the standard screening tool remains to be established.
- **Specificity:** Many neurodegeneration markers (e.g., NfL) are nonspecific to AD; distinguishing AD from other dementias remains a challenge.
- **Clinical implementation:** Translation from research to routine clinical practice, including establishing universal cut-off levels and standardized assays.
- **Atypical AD variants:** Biomarker profiles (especially amyloid imaging) are less tightly coupled to clinical presentation in atypical variants, complicating diagnosis.

Summary Table

Biomarker	Type	Pathological Target	Clinical Maturity
A β 42, A β 42/40 (CSF)	Fluid	Amyloid plaques	Established
A β 42/40 (plasma)	Fluid	Amyloid plaques	Emerging/Clinical use
P-tau181, P-tau217 (CSF/plasma)	Fluid	Tau tangles	Established/Emerging
NfL (plasma/CSF)	Fluid	Neurodegeneration (general)	Emerging
GFAP (plasma)	Fluid	Astroglial activation	Investigational
Amyloid PET	Imaging	Amyloid plaques	Established
Tau PET	Imaging	Tau tangles	Established
FDG-PET	Imaging	Synaptic/metabolic dysfunction	Established

Biomarker	Type	Pathological Target	Clinical Maturity
Structural MRI	Imaging	Atrophy	Established
Mitochondrial miRNAs	Fluid	Synaptic/mitochondrial	Investigational
DNA methylation	Fluid	Epigenetic changes	Investigational

The field is rapidly evolving, with blood-based biomarkers representing the most transformative recent development, potentially enabling population-scale screening and earlier intervention.